

QUANTUM ENTROPY CONTRACTION AND FACTORIZATION FROM HYPERCONTRACTIVITY

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ABSTRACT. We prove that hypercontractivity implies entropy contraction for a single quantum channel without assuming detailed balance. For primitive quantum Markov semigroups that are KMS-symmetric with respect to a faithful invariant state σ , we obtain the modified logarithmic Sobolev bound $\alpha_1 \geq \frac{\lambda}{(2+\log \|\sigma^{-1}\|_\infty)}$, where λ is the spectral gap. This removes the assumption of L_p -regularity previously needed to obtain this bound by comparison with the logarithmic Sobolev constant. As an application, we show that hypercontractivity of the average of two conditional expectations implies approximate tensorization of relative entropy.

1. INTRODUCTION

Quantitative convergence to equilibrium can be studied through either norm estimates or entropy inequalities. Hypercontractivity describes the improvement of weighted L_p norms under a Markov evolution, while entropy contraction measures the decrease of relative entropy. In this work, we study how hypercontractive estimates imply entropy contraction in finite-dimensional quantum systems. We note that the passage from hypercontractivity (HC) to entropy contraction (EC) already makes sense for a single channel. This is inspired by the work of Salez [20] in the classical setting, where neither reversibility nor regularity is assumed. In the quantum setting, the corresponding statement follows from Hirche–Rouzé–França [12, Proposition 5.4] through KMS duality. We combine it with a centering criterion that removes the multiplicative defect in a hyperbounded (HB) estimate using strict L_2 -contraction on centered elements.

Let σ be a faithful state, and let $E_\sigma(X) = \text{Tr}(\sigma X)I$ be the replacer map. The symmetric weighted norms are given by $\|X\|_{p,\sigma} = \|\sigma^{1/(2p)} X \sigma^{1/(2p)}\|_p$, and $D(\rho|\sigma) = \text{Tr}(\rho \log \rho - \rho \log \sigma)$ denotes the quantum relative entropy. We write Φ_* for the trace dual of a Heisenberg-picture channel Φ .

Theorem A (Single-channel contraction). *Let Φ be a unital completely positive map preserving σ .*

(i) *If $1 < p < q < \infty$ and $\|\Phi\|_{p \rightarrow q, \sigma} \leq 1$, then, for every state ρ ,*

$$D(\Phi_* \rho | \sigma) \leq \frac{p}{q} D(\rho | \sigma).$$

(ii) *If, for some $2 < q < \infty$, $M_q \geq 1$, and $0 < r < 1$,*

$$\|\Phi\|_{2 \rightarrow q, \sigma} \leq M_q, \quad \|\Phi - E_\sigma\|_{2 \rightarrow 2, \sigma} \leq r,$$

then $\|\Phi\|_{2 \rightarrow p, \sigma} \leq 1$ and, for every state ρ ,

$$D(\Phi_* \rho | \sigma) \leq \frac{2}{p} D(\rho | \sigma),$$

where

$$p = 2 + \frac{2(q-2) \log(1/r)}{q-2+q \log(M_q/r)} > 2.$$

Part (i) is established in Proposition 2.2 by applying [12, Proposition 5.4] and identifying the relevant KMS adjoint with the trace adjoint of the Petz map. Part (ii), proved in Theorem 2.4, is a weighted version of the centering method of Wang [25, Theorem 1.6]. In finite dimensions, a finite HB bound is automatic; its quantitative value determines the HC exponent supplied by

the centering method. These statements require neither symmetry nor a semigroup structure and provide the direct norm-to-entropy passage used below.

For a primitive quantum Markov semigroup (QMS) $P_t = e^{-t\mathcal{L}}$, we write α_1 , α_2 , and α_H for the constants associated with the modified logarithmic Sobolev inequality (MLSI), the L_2 -logarithmic Sobolev inequality (LSI), and HC, respectively; see Section 3.1 for detailed definitions. In the continuous-time setting, we obtain the following entropy contraction bounds.

Theorem B (MLSI from hyperboundedness). *Let (P_t) be a finite-dimensional primitive KMS-symmetric QMS with a faithful invariant state σ and spectral gap $\lambda > 0$. If, for some $2 < q \leq \infty$, $t_q > 0$, $M_q \geq 1$,*

$$\|P_{t_q}\|_{2 \rightarrow q, \sigma} \leq M_q,$$

then

$$\alpha_1 \geq \beta, \quad \alpha_2 \geq 2\beta, \quad \alpha_H \geq \beta, \quad \beta = \frac{(1 - 2/q)\lambda}{2[\lambda t_q + \log M_q + 1 - 2/q]}.$$

Moreover, for a primitive QMS with positive KMS coercivity constant $\lambda_{\text{KMS}} > 0$, without assuming symmetry,

$$\alpha_1 \geq \frac{\alpha_H}{2} \geq \frac{\lambda_{\text{KMS}}}{2 + \log \|\sigma^{-1}\|_\infty}. \quad (1.1)$$

We emphasize that no L_p -regularity is assumed in Theorem 3.3. This estimate is proved by constructing a local HC curve for small t from the endpoint estimate and the spectral gap. Its slope at $t = 0$ gives the MLSI and LSI bounds; extending the curve to all times gives the HC bound. The interpolation method, the existence of an all-time HC curve, and the implication from HC to LSI without L_p -regularity are already present in [22]. Here the centering criterion in Lemma 2.3 provides explicit constants, and the HC-to-EC implication gives MLSI directly. The quantitative comparison with [22] is given in Section 3.2.

For KMS-symmetric semigroups, bounds similar to (1.1) were obtained for the LSI constant α_2 , but the comparison with the MLSI constant has required L_p -regularity. Here, we remove this technical assumption and obtain the bound for all KMS-symmetric semigroups. Related qualitative MLSI results for primitive KMS-symmetric semigroups also recently appeared in [27, 28].

Let $E_i : \mathcal{M} \rightarrow \mathcal{N}_i$, $i = 1, 2$, be conditional expectations preserving a common faithful state σ , and let $E_{\mathcal{N}}$ be the corresponding expectation onto $\mathcal{N} = \mathcal{N}_1 \cap \mathcal{N}_2$. Write

$$\Phi_{\text{av}} = \frac{E_1 + E_2}{2}.$$

Approximate tensorization (AT) means that there is a constant $c > 0$ such that

$$D(\rho \| E_{\mathcal{N}*}\rho) \leq c(D(\rho \| E_{1*}\rho) + D(\rho \| E_{2*}\rho))$$

for all states ρ . When $\mathcal{N} \neq \mathcal{M}$, AT with constant $c > 0$ implies EC for $\Phi_{\text{av},*}$ with coefficient $1 - 1/(2c)$, by the chain rule and convexity [4, 8]. Our third theorem proves a converse whose AT constant depends only on the averaged EC coefficient.

Theorem C (AT from averaged EC). *Let $E_i : \mathcal{M} \rightarrow \mathcal{N}_i$, $i = 1, 2$, be conditional expectations preserving a common faithful state σ , and let $E_{\mathcal{N}}$ be the corresponding expectation onto $\mathcal{N} = \mathcal{N}_1 \cap \mathcal{N}_2$. If, for some $0 \leq \eta < 1$ and every state ρ ,*

$$D\left(\frac{E_{1*}\rho + E_{2*}\rho}{2} \| E_{\mathcal{N}*}\rho\right) \leq \eta D(\rho \| E_{\mathcal{N}*}\rho),$$

then

$$D(\rho \| E_{\mathcal{N}*}\rho) \leq \frac{3}{2(1-\eta)} (D(\rho \| E_{1*}\rho) + D(\rho \| E_{2*}\rho)). \quad (1.2)$$

This theorem makes averaged EC and AT quantitatively equivalent up to universal constants. The proof uses the quantum Jensen–Shannon divergence and the triangle inequality for the square root of this divergence [24]. Approximate tensorization for noncommuting conditional expectations and its complete versions were studied in [4, 11]. Gao and Rouzé [11, Corollary 5.4 and Remark 5.5] also obtain index-independent bounds from mixing in completely positive order, a condition of ultracontractive $L_1 \rightarrow L_\infty$ type; an index factor enters when one converts L_2 estimates into that mixing condition. Our result starts with an entropy contraction estimate and converts it to AT without an additional index factor. As an application of Theorem A, when $\mathcal{N} = \mathbb{C}I$, the hypercontractivity of the average channel $\rho \mapsto \frac{1}{2}(E_{1*}\rho + E_{2*}\rho)$ supplies the EC hypothesis needed to obtain AT.

Section 2 develops the single-channel estimates. Section 3 treats quantum Markov semigroups and discusses entropy decay from time zero as well as delayed estimates. Section 4 proves the implication from averaged EC to AT.

2. ENTROPY CONTRACTION VIA HYPERCONTRACTIVITY

We derive entropy contraction from hypercontractivity using a dual form of [12, Proposition 5.4]. Hypercontractivity can be obtained from a hyperbounded endpoint and strict L_2 contraction on centered observables, using a weighted version of the method in [25, Theorem 1.6]. The centered criterion will be used directly for semigroups in Section 3.

2.1. Weighted norms and state-preserving channels. Throughout the paper, let $B(\mathcal{H})$ denote the algebra of linear operators on a finite-dimensional Hilbert space $\mathcal{H}$ and let Tr denote the standard matrix trace. We identify states with density matrices and fix a faithful state $\sigma > 0$, $\text{Tr} \sigma = 1$. All logarithms are natural.

For $1 \leq p < \infty$, recall the symmetric weighted noncommutative L_p norms of [14, equations (10)–(12)], defined by

$$\|X\|_{p,\sigma} = \|\sigma^{1/(2p)} X \sigma^{1/(2p)}\|_p, \quad \|X\|_{\infty,\sigma} = \|X\|_\infty, \quad (2.1)$$

where the p -norm on the right is the Schatten p -norm. We write $L_p(\sigma)$ for this normed space, and $\|T\|_{p \rightarrow q,\sigma}$ for the induced operator norm; the subscript σ is omitted whenever there is no ambiguity. For $p = 2$, $L_2(\sigma)$ is a Hilbert space equipped with the Kubo–Martin–Schwinger (KMS) inner product

$$\langle X, Y \rangle_\sigma = \text{Tr}(X^* \sigma^{1/2} Y \sigma^{1/2}), \quad \Gamma_\sigma(X) = \sigma^{1/2} X \sigma^{1/2}, \quad (2.2)$$

where Γ_σ is the weighting map. For real s we write $\Gamma_\sigma^s(X) = \sigma^{s/2} X \sigma^{s/2}$. For conjugate exponents $1/p + 1/p' = 1$, weighted Hölder duality gives

$$|\langle X, Y \rangle_\sigma| \leq \|X\|_{p,\sigma} \|Y\|_{p',\sigma}, \quad \|X\|_{p,\sigma} = \sup_{\|Y\|_{p',\sigma} \leq 1} |\langle Y, X \rangle_\sigma|. \quad (2.3)$$

The norms are monotone: $\|X\|_{p,\sigma} \leq \|X\|_{q,\sigma}$ for $p \leq q$. The spaces $L_p(\sigma)$ form a complex interpolation scale [5, Theorem 4],

$$\|T\|_{p_\theta \rightarrow q_\theta} \leq \|T\|_{p_0 \rightarrow q_0}^{1-\theta} \|T\|_{p_1 \rightarrow q_1}^\theta, \quad \frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \quad \frac{1}{q_\theta} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}. \quad (2.4)$$

We will also use the analytic-family version of Riesz–Thorin interpolation in Section 3.

A quantum channel in the Heisenberg picture is a unital completely positive map $\Phi : B(\mathcal{H}) \rightarrow B(\mathcal{H})$. Its trace dual Φ_* in the Schrödinger picture is a completely positive trace-preserving (CPTP) map given by

$$\text{Tr}(\Phi_* \rho X) = \text{Tr}(\rho \Phi X).$$

Throughout the paper, we assume that Φ is σ -preserving, i.e. $\sigma = \Phi_* \sigma$ is an invariant state. Every σ -preserving channel is contractive

$$\|\Phi(X)\|_{p,\sigma} \leq \|X\|_{p,\sigma}$$

on $L_p(\sigma)$ for $1 \leq p \leq \infty$. The KMS adjoint with respect to $L_2(\sigma)$ is

$$\Phi^\# = \Gamma_\sigma^{-1} \Phi_* \Gamma_\sigma. \quad (2.5)$$

This is exactly the Petz recovery map of Φ_* with respect to the invariant state σ in the Heisenberg picture. In particular, $\Phi^\#$ is unital and completely positive and satisfies $\sigma \circ \Phi^\# = \sigma$. We call Φ KMS-symmetric if $\Phi^\# = \Phi$.

2.2. From norm estimates to entropy contraction. For two states ρ, ω , their Umegaki relative entropy [23] is

$$D(\rho \parallel \omega) = \text{Tr} \rho (\log \rho - \log \omega)$$

if $\text{supp } \rho \subseteq \text{supp } \omega$, and is $+\infty$ otherwise. We recall three functional inequalities for state-preserving quantum channels.

Definition 2.1 (Hypercontractivity, hyperboundedness and entropy contraction). Let $\Phi : B(\mathcal{H}) \rightarrow B(\mathcal{H})$ be a unital completely positive map and assume Φ preserves a fixed faithful state $\sigma = \Phi_* \sigma$. For $1 < p < q < \infty$, we say that

i) Φ is *hypercontractive* (HC) at (p, q) if

$$\|\Phi\|_{p \rightarrow q, \sigma} \leq 1.$$

ii) Φ is *hyperbounded* (HB) at (p, q) if for some $1 \leq M < \infty$,

$$\|\Phi\|_{p \rightarrow q, \sigma} \leq M.$$

iii) Φ satisfies *(relative) entropy contraction* with respect to σ if for some $0 < \eta < 1$,

$$D(\Phi_* \rho \parallel \sigma) \leq \eta D(\rho \parallel \sigma), \quad \forall \text{ state } \rho. \quad (2.6)$$

In finite dimensions, a hyperbounded estimate always holds and the information lies in the value of M and the dependence on the parameters (p, q) . Hypercontractivity is the case $M = 1$, which can hold only if Φ is primitive, i.e. Φ^k converges to E_σ . In particular, σ is the unique invariant state and the fixed-point space $\mathcal{N} := \{X \mid \Phi(X) = X\} = \mathbb{C}1$ is trivial. By the data processing inequality, the relative entropy is always monotone non-increasing

$$D(\Phi_* \rho \parallel \sigma) \leq D(\rho \parallel \sigma).$$

The entropy contraction (2.6) asserts a uniform contraction factor strictly smaller than one and is also called a strong data processing inequality (SDPI).

Since $\Phi_* \sigma = \sigma$, the Petz map of Φ_* at σ satisfies

$$R_{\sigma, \Phi_*} = \Gamma_\sigma \Phi \Gamma_\sigma^{-1}, \quad R_{\sigma, \Phi_*}^* = \Gamma_\sigma^{-1} \Phi_* \Gamma_\sigma = \Phi^\#. \quad (2.7)$$

Here $*$ denotes the trace adjoint. Weighted Hölder duality (2.3) gives

$$\|\Phi\|_{p \rightarrow q, \sigma} = \|\Phi^\#\|_{q' \rightarrow p', \sigma}, \quad \|\Phi^\#\|_{1 \rightarrow 1, \sigma} = 1. \quad (2.8)$$

Applying [12, Proposition 5.4] using the KMS duality above, we obtain a quantum analogue of Salez's result [20, Theorem 1] for classical Markov maps. No symmetry (reversibility) is required, as in the classical case [20].

Proposition 2.2 (Entropy contraction from hypercontractivity). *Let $1 < p \leq q < \infty$. Suppose Φ is a σ -preserving channel satisfying*

$$\|\Phi\|_{p \rightarrow q, \sigma} \leq 1. \quad (2.9)$$

Then, for every state ρ ,

$$D(\Phi_* \rho \parallel \sigma) \leq \frac{p}{q} D(\rho \parallel \sigma). \quad (2.10)$$

Proof. For $0 \leq t \leq 1$, set

$$a_t = (1 - t/q)^{-1}, \quad b_t = (1 - t/p)^{-1}.$$

By (2.8) and (2.9), complex interpolation yields

$$\|\Phi^\#\|_{a_t \rightarrow b_t, \sigma} \leq 1.$$

For a faithful state ρ , take $X = \Gamma_\sigma^{-1}\rho$. Then $\Phi^\#X = \Gamma_\sigma^{-1}\Phi_*\rho$, and both have unit weighted L_1 norm. The norm-entropy derivative [12, Lemma 5.1] is

$$\left. \frac{d}{ds} \log \|\Gamma_\sigma^{-1}\rho\|_{s, \sigma} \right|_{s=1} = D(\rho\|\sigma).$$

Taking logarithms and differentiating the interpolated estimate at $t = 0$ gives

$$\frac{1}{p} D(\Phi_*\rho\|\sigma) \leq \frac{1}{q} D(\rho\|\sigma),$$

which is (2.10). The result for a general state ρ follows by approximating it with the faithful states $(1 - \varepsilon)\rho + \varepsilon\sigma$. $\square$

Hypercontractivity also implies primitivity. Recall that a channel is primitive if its iterates Φ^k converge to the scalar expectation $E_\sigma(X) = \text{Tr}(\sigma X)I$. If $\|\Phi\|_{p \rightarrow q, \sigma} \leq 1$ with $p < q$, iteration gives

$$D(\Phi_*^k \rho\|\sigma) \leq \left(\frac{p}{q}\right)^k D(\rho\|\sigma) \rightarrow 0, \quad \text{as } k \rightarrow \infty. \quad (2.11)$$

By quantum Pinsker's inequality [14, Section I], this yields trace norm convergence

$$\|\Phi_*^k \rho - \sigma\|_1^2 \leq 2 \left(\frac{p}{q}\right)^k D(\rho\|\sigma).$$

When the fixed-point algebra $\mathcal{N}$ is nontrivial, relative hyperboundedness was introduced in [3] using amalgamated norms relative to $\mathcal{N}$. If E is the fixed-point conditional expectation and $\Phi E = E\Phi = E$, the corresponding entropy contraction is formulated using $D(\rho\|E_*\rho)$, the decoherence-free relative entropy introduced in [2, Definition 3.3], whose reference state $E_*\rho$ depends on the input ρ . However, for the amalgamated-norm formulation of [3], a nontrivially decohering QMS on $B(\mathcal{H})$ cannot satisfy a defect-free HC curve with positive initial slope when $\mathbb{C}I \subsetneq \mathcal{N} \subsetneq B(\mathcal{H})$; see [3, Theorems 2.4(i) and 5.1].

2.3. From hyperboundedness to hypercontractivity. It follows by an argument similar to that of Proposition 2.2 that hyperboundedness implies entropy contraction with a defect

$$\|\Phi\|_{p \rightarrow q, \sigma} \leq M \quad \implies \quad D(\Phi_*\rho\|\sigma) \leq \frac{p}{q} D(\rho\|\sigma) + p \log M.$$

We now use strict L_2 coercivity to remove the defect term. The key ingredient is an L_p -norm analogue of the Rothaus lemma, based on a centering argument. We state the centering argument separately, as it will be reused in Section 3.

Define the expectation map

$$E_\sigma(X) = \text{Tr}(\sigma X)I. \quad (2.12)$$

This gives the KMS orthogonal decomposition $L_2(\sigma) = \mathbb{C}I \oplus \ker E_\sigma$, with

$$\|aI + X_0\|_{2, \sigma}^2 = |a|^2 + \|X_0\|_{2, \sigma}^2$$

for $X_0 \in \ker E_\sigma$. Every σ -preserving channel satisfies $\Phi E_\sigma = E_\sigma \Phi = E_\sigma$, and therefore

$$\|\Phi - E_\sigma\|_{2 \rightarrow 2, \sigma} = \|\Phi|_{\ker E_\sigma}\|_{2 \rightarrow 2, \sigma}. \quad (2.13)$$

Lemma 2.3. *Let Φ be a σ -preserving channel and $2 \leq p < \infty$. If*

$$\|\Phi - E_\sigma\|_{2 \rightarrow p, \sigma} \leq (p - 1)^{-1/2}, \quad (2.14)$$

then $\|\Phi\|_{2 \rightarrow p, \sigma} \leq 1$.

Proof. Lemma 2.9 of [29], in its reversed form for $p > 2$, gives

$$\|X\|_{p,\sigma}^2 \leq |\text{Tr}(\sigma X)|^2 + (p-1)\|X - E_\sigma X\|_{p,\sigma}^2. \quad (2.15)$$

Writing $X = aI + X_0$, with $X_0 \in \ker E_\sigma$, we obtain

$$\|\Phi X\|_{p,\sigma}^2 \leq |a|^2 + (p-1)\|\Phi X_0\|_{p,\sigma}^2 \leq |a|^2 + \|X_0\|_{2,\sigma}^2 = \|X\|_{2,\sigma}^2. \quad \square$$

The next theorem is a weighted version of [25, Theorem 1.6]. It upgrades a hyperbounded estimate to hypercontractivity via centered L_2 contraction and hence gives entropy contraction.

Theorem 2.4. *Let Φ be a σ -preserving channel. Suppose that*

i) *for some $2 < q < \infty$ and $M_q \geq 1$,*

$$\|\Phi\|_{2 \rightarrow q,\sigma} \leq M_q, \quad (2.16)$$

ii) *for some $0 < r < 1$,*

$$\|\Phi - E_\sigma\|_{2 \rightarrow 2,\sigma} \leq r, \quad (2.17)$$

Then

$$\|\Phi\|_{2 \rightarrow p,\sigma} \leq 1, \quad (2.18)$$

and for every state ρ ,

$$D(\Phi_*\rho\|\sigma) \leq \frac{2}{p}D(\rho\|\sigma) \quad (2.19)$$

where

$$p = 2 + \frac{2(q-2)\log(1/r)}{q-2+q\log(M_q/r)}. \quad (2.20)$$

Proof. For $0 < r < 1$ and $M_q \geq 1$, the definition of p gives $0 < p-2 < 2(q-2)/q < q-2$. For $2 < p \leq q$, put $\theta_p = (1/2 - 1/p)/(1/2 - 1/q)$. For X with $E_\sigma(X) = 0$, interpolation of the output norms gives

$$\|\Phi X\|_{p,\sigma} \leq \|\Phi X\|_{2,\sigma}^{1-\theta_p} \|\Phi X\|_{q,\sigma}^{\theta_p} \leq r^{1-\theta_p} M_q^{\theta_p} \|X\|_{2,\sigma}.$$

Thus Lemma 2.3 gives hypercontractivity whenever

$$(p-1)r^{2(1-\theta_p)}M_q^{2\theta_p} \leq 1. \quad (2.21)$$

By the inequalities $\log(p-1) \leq p-2$ and $\theta_p \leq q(p-2)/(2(q-2))$, the logarithm of the left-hand side is bounded by

$$\log\left((p-1)r^{2(1-\theta_p)}M_q^{2\theta_p}\right) \leq 2\log r + \left(1 + \frac{q}{q-2}\log(M_q/r)\right)(p-2) \leq 0,$$

where p is chosen as in (2.20) to ensure the last inequality. This proves hypercontractivity at $(2, p)$, which gives entropy contraction (2.19) by Proposition 2.2. $\square$

Remark 2.5. a) For a primitive channel Φ that is KMS-symmetric with respect to σ , condition ii) (2.17) is always satisfied. Indeed, under KMS symmetry, Φ is a self-adjoint operator on $L_2(\sigma)$ commuting with E_σ , and hence the norm of $\Phi - E_\sigma$ equals its spectral radius, which is strictly less than 1 by primitivity.

b) The above qualitative implication from strict centered L_2 contraction to a nontrivial HC exponent is known in the classical setting [1, Theorem 3(a), (c), p. 928]. In finite-dimensional noncommutative algebras with a tracial state, Wang [25, Theorem 1.6] proves the corresponding $s \rightarrow 2$ characterization for $1 < s < 2$. Our argument above is a weighted version in the dual range using the same p -convexity inequality.

3. QUANTUM MARKOV SEMIGROUPS

In this section, we discuss how a hypercontractive or hyperbounded estimate implies entropy contraction in continuous time.

3.1. Functional inequalities. A *quantum Markov semigroup* (QMS) is a continuous semigroup $(P_t)_{t \geq 0} : B(\mathcal{H}) \rightarrow B(\mathcal{H})$ of unital completely positive maps satisfying $P_t \circ P_s = P_{t+s}$ for $s, t \geq 0$ and $P_0 = \text{id}$. Throughout this section, we assume $\mathcal{H}$ is finite-dimensional, and the semigroup is primitive with a faithful invariant state σ . We write $P_t = e^{-t\mathcal{L}}$ with the generator

$$\mathcal{L}X = -\lim_{t \rightarrow 0} \frac{1}{t}(P_t X - X),$$

and $K = \ker E_\sigma$. KMS symmetry is assumed only where stated, and we omit σ from p -norms when it is clear from the context. For a faithful state ρ , the entropy production is [14, equations (5)–(6)]

$$\mathcal{I}_\mathcal{L}(\rho) = \text{Tr}[(\mathcal{L}_*\rho)(\log \rho - \log \sigma)] = -\left. \frac{d}{dt} D(P_{t*}\rho \| \sigma) \right|_{t=0}. \quad (3.1)$$

The HC constant α_H is the supremum of $\alpha \geq 0$ such that

$$\|P_t\|_{2 \rightarrow 1 + e^{2\alpha t}, \sigma} \leq 1 \quad (t \geq 0). \quad (3.2)$$

For $X > 0$, write $Y = \Gamma_{\frac{1}{2}}^{\frac{1}{2}}(X) = \sigma^{\frac{1}{4}} X \sigma^{\frac{1}{4}}$, and set

$$\text{Ent}_{2,\sigma}(X) = \frac{1}{2} \{ \text{Tr}[Y^2(\log Y^2 - \log \sigma)] - \text{Tr}(Y^2) \log \text{Tr}(Y^2) \}, \quad \mathcal{E}(X) = \text{Re} \langle X, \mathcal{L}X \rangle_\sigma.$$

Define the constants α_2 and α_1 as the best nonnegative constants in the L_2 -logarithmic Sobolev inequality (LSI) and the modified logarithmic Sobolev inequality (MLSI), respectively:

$$\alpha_2 \text{Ent}_{2,\sigma}(X) \leq \mathcal{E}(X), \quad (\text{LSI})$$

$$2\alpha_1 D(\rho \| \sigma) \leq \mathcal{I}_\mathcal{L}(\rho). \quad (\text{MLSI})$$

By Gronwall's lemma, MLSI is equivalent to exponential entropy decay from time zero [14, equations (5)–(6)]:

$$D(P_{t*}\rho \| \sigma) \leq e^{-2\alpha_1 t} D(\rho \| \sigma), \quad t \geq 0. \quad (3.3)$$

We also use the norm–entropy derivative [14, Theorem 4]:

$$\partial_r \log \|X\|_{r,\sigma} \big|_{r=2} = \frac{\text{Ent}_{2,\sigma}(X)}{2\|X\|_{2,\sigma}^2}, \quad X > 0, \quad (3.4)$$

which gives $\alpha_H \leq \alpha_2$ [14, Theorem 15(1)]. Direct Dirichlet-form comparison under KMS symmetry gives $\alpha_1 \geq \alpha_2$ under strong L_p -regularity [14, Propositions 8 and 13], a condition satisfied by GNS-symmetric semigroups but not known to follow from KMS symmetry. In summary, for a primitive KMS-symmetric semigroup

$$\alpha_H \leq \alpha_2 \stackrel{(*)}{\leq} \alpha_1$$

where $(*)$ requires strong L_p -regularity.

The next lemma shows that $\alpha_H \leq \alpha_2$ and $\alpha_H \leq 2\alpha_1$ without KMS symmetry or L_p -regularity. For KMS-symmetric semigroups, the argument from HC to MLSI was discussed in [12, Section 7, EQ. (7.2)–(7.3)].

Lemma 3.1 (Local HC curve). *Let $r : [0, \varepsilon) \rightarrow [2, \infty)$ be right differentiable at zero, with $r(0) = 2$ and $r'(0) = v > 0$. If $\|P_s\|_{2 \rightarrow r(s)} \leq 1$ for $0 \leq s < \varepsilon$, then*

$$\alpha_1 \geq \frac{v}{4}, \quad \alpha_2 \geq \frac{v}{2}, \quad \alpha_H \geq \frac{v}{4}.$$

In particular, we have $\alpha_H \leq 2\alpha_1$ and $\alpha_H \leq \alpha_2$.

Proof. Proposition 2.2 gives

$$D(P_{s*}\rho\|\sigma) \leq \frac{2}{r(s)} D(\rho\|\sigma).$$

Since $2/r(s) = 1 - vs/2 + o(s)$, differentiation at zero gives $\alpha_1 \geq v/4$. For the LSI constant, the norm-entropy derivative (3.4) gives

$$0 \geq \frac{d}{ds} \log \|P_s X\|_{r(s),\sigma} \Big|_{s=0+} = \frac{(v/2) \text{Ent}_{2,\sigma}(X) - \mathcal{E}(X)}{\|X\|_{2,\sigma}^2}.$$

Thus $v \text{Ent}_{2,\sigma}(X) \leq 2\mathcal{E}(X)$, which gives $\alpha_2 \geq v/2$. For the all-time estimate, interpolation with $\|P_s\|_{\infty \rightarrow \infty} = 1$ gives

$$\|P_s\|_{p \rightarrow pr(s)/2} \leq 1, \quad 2 \leq p < \infty.$$

For fixed $t > 0$, composition over n steps of length t/n yields

$$\|P_t\|_{2 \rightarrow 2(r(t/n)/2)^n} \leq 1,$$

where the target exponent converges to $2e^{vt/2}$ as $n \rightarrow \infty$. The bound on α_H follows from $1 + e^{vt/2} \leq 2e^{vt/2}$ and norm monotonicity. $\square$

3.2. Modified logarithmic Sobolev bounds. For a finite-dimensional primitive QMS $P_t = e^{-\mathcal{L}t}$ with $\lim_{t \rightarrow \infty} P_t = E_\sigma$, we define the KMS coercivity constant

$$\lambda_{\text{KMS}} := \inf_{0 \neq X \in \ker E_\sigma} \frac{\text{Re}\langle X, \mathcal{L}X \rangle_\sigma}{\|X\|_{2,\sigma}^2} \geq 0.$$

Under KMS symmetry, $\mathcal{L}$ is self-adjoint on $L_2(\sigma)$ and λ_{KMS} is exactly the spectral gap

$$\lambda_{\text{KMS}} = \inf_{0 \neq X \in \ker E_\sigma} \frac{\langle X, \mathcal{L}X \rangle_\sigma}{\|X\|_{2,\sigma}^2} = \min(\text{spec}(\mathcal{L}) \setminus \{0\}) =: \lambda > 0, \quad (3.5)$$

which is always positive in finite dimensions, and one has [14, Theorem 16]

$$\alpha_1 \leq \lambda. \quad (3.6)$$

For a finite-dimensional QMS without KMS symmetry, one has $\alpha_1 \leq 2\lambda_{\text{KMS}}$, with a factor of 2 lost through symmetrization.

We derive the modified logarithmic Sobolev inequality from hyperboundedness using the coercivity constant or spectral gap on centered elements. The argument applies with or without KMS symmetry and does not require L_p -regularity.

Theorem 3.2 (Bounds from centered coercivity). *Let (P_t) be a σ -preserving primitive QMS with a positive KMS coercivity constant $\lambda_{\text{KMS}} = \lambda > 0$. Then*

$$\alpha_1 \geq \frac{\alpha_H}{2} \geq \frac{\lambda}{2 + \log \|\sigma^{-1}\|_\infty}. \quad (3.7)$$

Proof. For $X \in \ker E_\sigma$, coercivity gives

$$\frac{d}{dt} \|P_t X\|_{2,\sigma}^2 \leq -2\lambda \|P_t X\|_{2,\sigma}^2,$$

which implies $\|P_t - E_\sigma\|_{2 \rightarrow 2} \leq e^{-\lambda t}$. The elementary weighted norm comparison is

$$\|X\|_{r,\sigma} \leq \|\sigma^{-1}\|_\infty^{1/2-1/r} \|X\|_{2,\sigma}, \quad 2 \leq r < \infty. \quad (3.8)$$

Indeed, put $Z = \sigma^{1/4} X \sigma^{1/4}$ and $B = \sigma^{1/(2r)-1/4}$. The ideal property and Schatten norm monotonicity give

$$\begin{aligned} \|X\|_{r,\sigma} &= \|\sigma^{1/(2r)-1/4} (\sigma^{1/4} X \sigma^{1/4}) \sigma^{1/(2r)-1/4}\|_r \leq \|\sigma^{1/(2r)-1/4}\|_\infty^2 \|\sigma^{1/4} X \sigma^{1/4}\|_r \\ &\leq \|\sigma^{-1}\|_\infty^{1/2-1/r} \|\sigma^{1/4} X \sigma^{1/4}\|_2 = \|\sigma^{-1}\|_\infty^{1/2-1/r} \|X\|_{2,\sigma} \end{aligned}$$

The case $r = 4$ is used in [22, Corollary 6]. Thus

$$\|P_t - E_\sigma\|_{2 \rightarrow r} \leq \|\sigma^{-1}\|_\infty^{1/2-1/r} e^{-\lambda t}.$$

Lemma 2.3 and Proposition 2.2 give HC $2 \rightarrow r$ and EC with coefficient $2/r$ whenever

$$t \geq \frac{\log(r-1) + (1-2/r) \log \|\sigma^{-1}\|_\infty}{2\lambda}. \quad (3.9)$$

For $r(t) = 1 + e^{2\alpha t}$, this condition becomes

$$(\lambda - \alpha)t \geq \frac{\log \|\sigma^{-1}\|_\infty}{2} \tanh(\alpha t). \quad (3.10)$$

The elementary bound $\tanh(\alpha t) \leq \alpha t$ shows that (3.10) holds for every $t \geq 0$ whenever

$$0 \leq \alpha \leq A := \frac{2\lambda}{2 + \log \|\sigma^{-1}\|_\infty}.$$

In particular, choosing $\alpha = A$ gives

$$\|P_t\|_{2 \rightarrow 1+e^{2At}, \sigma} \leq 1 \quad (t \geq 0).$$

By the definition of α_H in (3.2), this implies $\alpha_H \geq A$. Together with $\alpha_1 \geq \alpha_H/2$ from Lemma 3.1, this proves (3.7). $\square$

The above theorem gives HC and entropy decay using only coercivity and the smallest eigenvalue of σ , without KMS symmetry or L_p -regularity. A similar estimate for the LSI constant α_2 was obtained in [22] for KMS-symmetric semigroups through Rothaus's lemma. Using this estimate to bound the MLSI constant α_1 would require L_p -regularity. In particular, our bound (3.7) improves the qualitative MLSI conclusion for primitive KMS-symmetric semigroups that also appears in [27, Theorem 3.1] and [28, Corollary 6.1]. Moreover, the logarithmic loss $\log \|\sigma^{-1}\|_\infty$ is asymptotically tight, as in the classical case.

A faster curve is available after a delay: for $0 < \alpha < \lambda$, use $\tanh(\alpha t) \leq 1$ in (3.10) to similarly obtain the HC and EC estimates whenever

$$t \geq \frac{\log \|\sigma^{-1}\|_\infty}{2(\lambda - \alpha)}. \quad (3.11)$$

For the tracial state $\sigma = I/d$ on M_d , a stronger reversible comparison is recalled in [17, Theorem 2.2]:

$$\alpha_1 \geq \alpha_2 \geq \frac{2(1-2/d)\lambda}{\log(d-1)},$$

with coefficient 1 at $d = 2$. Thus (3.7) is a general faithful-state bound, not an optimal tracial bound.

The norm comparison at time zero $t = 0$ can be replaced by a single HB estimate. The interpolation argument below follows [22, Theorem 5] and the classical argument of [9, Theorem 3.9]. Combining it with the centered HC criterion gives the following bounds.

Theorem 3.3 (MLSI from a hyperbounded estimate). *Let P_t be a primitive QMS that is KMS-symmetric with respect to σ , and let $\lambda > 0$ be its spectral gap. Assume that for some $2 < q \leq \infty$, $t_q > 0$, and $M_q \geq 1$,*

$$\|P_{t_q}\|_{2 \rightarrow q} \leq M_q. \quad (3.12)$$

Then

$$\alpha_1 \geq \beta := \frac{(1-2/q)\lambda}{2[\lambda t_q + \log M_q + 1 - 2/q]}. \quad (3.13)$$

Moreover, the LSI and HC constants satisfy

$$\alpha_2 \geq 2\beta, \quad \alpha_H \geq \beta. \quad (3.14)$$

Proof. Set $b_q = 1 - 2/q$. We use the Stein–Weiss argument of [22, equations (18)–(19)]. For $K_z = e^{-zt_q\mathcal{L}}$ on the strip $0 \leq \operatorname{Re} z \leq 1$, KMS self-adjointness gives $\|K_{iy}\|_{2 \rightarrow 2} = 1$ and $\|K_{1+iy}\|_{2 \rightarrow q} \leq M_q$. Interpolation in the weighted L_p scale therefore yields

$$\|P_u\|_{2 \rightarrow p(u)} \leq M_q^{u/t_q}, \quad p(u) = \frac{2}{1 - b_q u/t_q}, \quad 0 \leq u \leq t_q. \quad (3.15)$$

For $2 \leq r \leq q$ with $r < \infty$, put

$$\theta_r = \frac{1 - 2/r}{b_q}, \quad u_r = t_q \theta_r. \quad (3.16)$$

Since $\|P_T|_{\ker E_\sigma}\|_{2 \rightarrow r} \leq M_q^{\theta_r} e^{-\lambda(T - u_r)}$ for $T \geq u_r$, Lemma 2.3 gives HC as soon as

$$T \geq F(r) := t_q \theta_r + \frac{\theta_r \log M_q + \frac{1}{2} \log(r - 1)}{\lambda}. \quad (3.17)$$

Together with Proposition 2.2, this proves the finite-exponent estimates

$$\|P_T\|_{2 \rightarrow r} \leq 1, \quad D(P_{T*}\rho|\sigma) \leq \frac{2}{r} D(\rho|\sigma). \quad (3.18)$$

The function F is continuously differentiable and strictly increasing on $[2, q]$, with $F(2) = 0$ and

$$F'(2) = \frac{\lambda t_q + \log M_q + b_q}{2b_q \lambda}.$$

Its inverse $r(T) = F^{-1}(T)$ is a local HC curve, defined for $T \geq 0$ sufficiently small, whose initial slope is $r'(0) = 4\beta$. Lemma 3.1 proves all three bounds. $\square$

The MLSI and LS_2 estimates use the initial slope directly; the extension to a full HC curve gives the rate β . In the normalization of [22], Theorems 3.2 and 3.3 give $\alpha_2 \geq 2\lambda/(2 + \log \|\sigma^{-1}\|_\infty)$ and $\alpha_2 \geq 2\beta$, respectively. These are twice the corresponding bounds in [22, Eqs. (30), (11)], after reversing the generator sign. The interpolation method and the existence of an all-time HC curve without L_p -regularity are already present in [22].

3.3. Delayed hypercontractivity and the lack of MLSI. Theorems 3.2 and 3.3 use L_2 -decay of centered elements with prefactor one. In finite dimensions, primitivity instead only guarantees an estimate

$$\|P_t - E_\sigma\|_{2 \rightarrow 2} \leq C e^{-\nu t}, \quad t \geq 0, \quad (3.19)$$

for some $C \geq 1$ and $\nu > 0$. Indeed, every positive rate below the smallest real part of the spectrum of $\mathcal{L}|_K$ is admissible after absorbing the Jordan polynomial factors into C . This does not ensure $C = 1$, and hence does not give the coercivity required in Theorem 3.2.

The norm comparison (3.8) gives

$$\|P_t - E_\sigma\|_{2 \rightarrow r} \leq C \|\sigma^{-1}\|_\infty^{1/2 - 1/r} e^{-\nu t}, \quad 2 \leq r < \infty.$$

For $r(t) = 1 + e^{2\alpha t}$, Lemma 2.3 gives HC whenever

$$(\nu - \alpha)t \geq \log C + \frac{1}{2} \log \|\sigma^{-1}\|_\infty \tanh(\alpha t).$$

Since $\tanh(\alpha t) \leq 1$, for every $0 < \alpha < \nu$ and

$$t \geq \tau(\alpha) := \frac{\log C + \frac{1}{2} \log \|\sigma^{-1}\|_\infty}{\nu - \alpha}, \quad (3.20)$$

we have

$$\|P_t\|_{2 \rightarrow 1 + e^{2\alpha t}} \leq 1, \quad D(P_{t*}\rho|\sigma) \leq \frac{2}{1 + e^{2\alpha t}} D(\rho|\sigma) \leq e^{-\alpha t} D(\rho|\sigma). \quad (3.21)$$

Here the entropy estimate follows from Proposition 2.2 and $2/(1 + e^{2\alpha t}) = e^{-\alpha t}/\cosh(\alpha t)$. Nevertheless, such a delayed estimate does not imply a positive MLSI constant $\alpha_1 > 0$: it does not provide entropy decay from time zero.

The following qubit model makes this limitation explicit. It appears in [10, section ‘‘Hypocoercivity and Physical Intuition’’]; the same generator and zero-entropy-production states occur in [27, Propositions 4.1–4.2].

Example 3.4 (Delayed decay with zero MLSI constant). Let $\sigma = I/2$ on M_2 and $h, \gamma > 0$. The QMS with generator

$$-\mathcal{L}_*(\rho) = -i[h\sigma_x, \rho] + \gamma(\sigma_z\rho\sigma_z - \rho)$$

is primitive and satisfies (3.19) for suitable C, ν , and therefore obeys the delayed HC and EC estimates (3.21), but $\alpha_1 = 0$ and $\lambda_{\text{KMS}} = 0$.

To verify these assertions, write $\rho = (I + x\sigma_x + y\sigma_y + z\sigma_z)/2$. The Bloch equations are

$$\dot{x} = -2\gamma x, \quad \frac{d}{dt} \begin{pmatrix} y \\ z \end{pmatrix} = \begin{pmatrix} -2\gamma & -2h \\ 2h & 0 \end{pmatrix} \begin{pmatrix} y \\ z \end{pmatrix}.$$

The two eigenvalues of the displayed matrix are $-\gamma \pm \sqrt{\gamma^2 - 4h^2}$ and have strictly negative real parts, so every state converges to $I/2$. For every $0 < \nu < \gamma - \text{Re} \sqrt{\gamma^2 - 4h^2}$, matrix exponential estimates give a finite $C \geq 1$ in (3.19), including the possible Jordan case.

For $\rho_a = (I + a\sigma_z)/2$, $0 < |a| < 1$, the dissipative part vanishes and the Hamiltonian part has zero entropy production. Hence

$$\mathcal{I}_{\mathcal{L}}(\rho_a) = 0, \quad D(\rho_a \| I/2) > 0,$$

which gives $\alpha_1 = 0$. Finally, $\text{Re}\langle \sigma_z, \mathcal{L}\sigma_z \rangle_\sigma = 0$ and L_2 contractivity imply $\lambda_{\text{KMS}} = 0$.

4. ENTROPY CONTRACTION AND APPROXIMATE TENSORIZATION

For two conditional expectations, approximate tensorization easily implies entropy contraction of their average. Theorem 4.1 proves that the converse also holds up to an absolute constant. The proof controls the Jensen–Shannon term in the entropy loss.

Let $E_i : B(\mathcal{H}) \rightarrow \mathcal{N}_i$, $i = 1, 2$, be conditional expectations preserving a common faithful state σ . Namely, each E_i is a unital completely positive map satisfying

$$E_i(AXB) = AE_i(X)B \quad \forall \quad A, B \in \mathcal{N}_i \quad \text{and} \quad E_{i*}\sigma = \sigma.$$

Each expectation is uniquely determined by the state σ it preserves and its range $\mathcal{N}_i$, and acts as an orthogonal projection in $L_2(\sigma)$ [4, Definition 1 and Proposition 1]. The intersection range $\mathcal{N} = \mathcal{N}_1 \cap \mathcal{N}_2$ also admits a σ -preserving expectation $E_{\mathcal{N}}$. In this section, we focus on the average of the two conditional expectations

$$\Phi_{\text{av}} = \frac{E_1 + E_2}{2}$$

as a positive self-adjoint contraction with fixed space $\mathcal{N}$. Its powers Φ_{av}^n converge to $E_{\mathcal{N}}$.

For a state ρ , write

$$\rho_i = E_{i*}\rho, \quad \rho_{\mathcal{N}} = E_{\mathcal{N}*}\rho, \quad D_{\mathcal{N}}(\rho) = D(\rho \| \rho_{\mathcal{N}}). \quad (4.1)$$

The inclusion $\mathcal{N} \subset \mathcal{N}_i$ and the chain rule imply that

$$E_{\mathcal{N}}E_i = E_{\mathcal{N}} = E_iE_{\mathcal{N}}, \quad D(\rho \| \rho_{\mathcal{N}}) = D(\rho \| \rho_i) + D(\rho_i \| \rho_{\mathcal{N}}), \quad i = 1, 2, \quad (4.2)$$

for all states ρ , including singular states [13, Lemma 3.4 of the arXiv version]. It is known that the exact entropy tensorization inequality (also called entropy factorization)

$$D(\rho \| \rho_{\mathcal{N}}) \leq D(\rho \| \rho_1) + D(\rho \| \rho_2)$$

holds for every state ρ if and only if the conditional expectations satisfy the commuting square condition

$$E_1 E_2 = E_2 E_1 = E_{\mathcal{N}}.$$

We use the defect-free approximate tensorization inequality of [4, Definition 2, with defect $d = 0$], but allow any constant $c > 0$ rather than imposing $c \geq 1$. We say that (E_1, E_2) satisfies approximate tensorization with constant $c > 0$, denoted by $\text{AT}(c)$, if

$$D(\rho \| \rho_{\mathcal{N}}) \leq c(D(\rho \| \rho_1) + D(\rho \| \rho_2)) \quad \text{for every state } \rho. \quad (4.3)$$

The optimal coefficient c_{AT} is the infimum of all admissible constants $c > 0$, with $c_{\text{AT}} = +\infty$ if none exists. Thus c_{AT} is the sharp coefficient in (4.3) and may be smaller than one. For example, if $E_1 = E_2 = E_{\mathcal{N}} \neq \text{id}$, then $c_{\text{AT}} = 1/2$. If $\mathcal{N} = B(\mathcal{H})$, both expectations are the identity and $c_{\text{AT}} = 0$; the formulas below involving entropy ratios or $1/c_{\text{AT}}$ are used only when $\mathcal{N} \neq B(\mathcal{H})$.

The averaged map is

$$\Phi_{\text{av}} = \frac{E_1 + E_2}{2}, \quad \Phi_{\text{av},*}\rho = \frac{\rho_1 + \rho_2}{2}, \quad (4.4)$$

Entropy contraction for the average means

$$D(\Phi_{\text{av},*}\rho \| \rho_{\mathcal{N}}) \leq \eta D(\rho \| \rho_{\mathcal{N}}) \quad \text{for every state } \rho, \quad 0 \leq \eta < 1, \quad (4.5)$$

and is denoted by $\text{EC}_{\mathcal{N}}(\eta)$. The reference state on both sides is $\rho_{\mathcal{N}}$, which depends on the input ρ .

The known implication from AT to average EC follows from (4.2) and convexity [8, Eqs. (11)–(12)]:

$$D(\Phi_{\text{av},*}\rho \| \rho_{\mathcal{N}}) \leq \frac{1}{2} \sum_{i=1}^2 D(\rho_i \| \rho_{\mathcal{N}}) \leq \left(1 - \frac{1}{2c_{\text{AT}}}\right) D(\rho \| \rho_{\mathcal{N}}). \quad (4.6)$$

The converse requires an upper bound on the entropy lost in the averaging step.

Theorem 4.1 (AT from EC). *Let $E_i : B(\mathcal{H}) \rightarrow \mathcal{N}_i$, $i = 1, 2$, be conditional expectations preserving a common faithful state σ , and let $E_{\mathcal{N}}$ be the σ -preserving conditional expectation onto the intersection $\mathcal{N} = \mathcal{N}_1 \cap \mathcal{N}_2$. Assume that the averaged map satisfies the following entropy contraction inequality for some $0 < \eta < 1$ and all states ρ :*

$$D(\Phi_{\text{av},*}\rho \| \rho_{\mathcal{N}}) \leq \eta D(\rho \| \rho_{\mathcal{N}}). \quad (4.7)$$

Then the following AT inequality holds:

$$D(\rho \| \rho_{\mathcal{N}}) \leq \frac{3}{2(1-\eta)} (D(\rho \| \rho_1) + D(\rho \| \rho_2)) \quad \text{for every state } \rho. \quad (4.8)$$

In particular,

$$c_{\text{AT}} \leq \frac{3}{2(1-\eta)}. \quad (4.9)$$

Proof. Set $\bar{\rho} = (\rho_1 + \rho_2)/2$. For states A, B , recall the quantum Jensen–Shannon divergence [24, Section 1.2]

$$J(A, B) = \frac{1}{2} D\left(A \left\| \frac{A+B}{2}\right.\right) + \frac{1}{2} D\left(B \left\| \frac{A+B}{2}\right.\right). \quad (4.10)$$

Donald's identity [21, equation (5)] reads

$$\frac{1}{2} D(A \| R) + \frac{1}{2} D(B \| R) = D\left(\frac{A+B}{2} \left\| R\right.\right) + J(A, B), \quad (4.11)$$

when $\text{supp } A, \text{supp } B \subseteq \text{supp } R$. Taking $A = \rho_1$, $B = \rho_2$, and $R = \rho_{\mathcal{N}}$, and using (4.2) gives the entropy-loss identity

$$D(\rho \| \rho_{\mathcal{N}}) - D(\bar{\rho} \| \rho_{\mathcal{N}}) = \frac{1}{2} (D(\rho \| \rho_1) + D(\rho \| \rho_2)) + J(\rho_1, \rho_2). \quad (4.12)$$

By Virosztek's metric theorem [24, Theorem 1], for all states A, B, C ,

$$\sqrt{J(A, C)} \leq \sqrt{J(A, B)} + \sqrt{J(B, C)}. \quad (4.13)$$

Taking $R = B$ in (4.11) and using the nonnegativity of relative entropy, we also have

$$J(A, B) \leq \frac{1}{2}D(A\|B), \quad J(A, B) \leq \frac{1}{2}D(B\|A). \quad (4.14)$$

Using the original input ρ as the intermediate state in (4.13) yields

$$\sqrt{J(\rho_1, \rho_2)} \leq \sqrt{J(\rho_1, \rho)} + \sqrt{J(\rho, \rho_2)} \leq \sqrt{\frac{D(\rho\|\rho_1)}{2}} + \sqrt{\frac{D(\rho\|\rho_2)}{2}}.$$

Combining the above estimates with the entropy contraction assumption (4.7), we obtain

$$(1 - \eta)D(\rho\|\rho_{\mathcal{N}}) \leq D(\rho\|\rho_{\mathcal{N}}) - D(\bar{\rho}\|\rho_{\mathcal{N}}) \quad (4.15)$$

$$\leq \frac{1}{2}(D(\rho\|\rho_1) + D(\rho\|\rho_2)) + J(\rho_1, \rho_2) \quad (4.16)$$

$$\leq \frac{1}{2}(D(\rho\|\rho_1) + D(\rho\|\rho_2)) + \frac{1}{2}\left(\sqrt{D(\rho\|\rho_1)} + \sqrt{D(\rho\|\rho_2)}\right)^2 \quad (4.17)$$

$$= D(\rho\|\rho_1) + D(\rho\|\rho_2) + \sqrt{D(\rho\|\rho_1)D(\rho\|\rho_2)} \quad (4.18)$$

$$\leq \frac{3}{2}(D(\rho\|\rho_1) + D(\rho\|\rho_2)), \quad (4.19)$$

where the first inequality follows from entropy contraction $D(\bar{\rho}\|\rho_{\mathcal{N}}) \leq \eta D(\rho\|\rho_{\mathcal{N}})$. This completes the proof. $\square$

The above bound holds for an arbitrary intersection $\mathcal{N}_1 \cap \mathcal{N}_2 = \mathcal{N}$. It also holds completely: a complete EC bound with one coefficient $\eta < 1$ implies complete AT with the same constant $3/[2(1-\eta)]$. Approximate tensorization for noncommuting expectations and its complete versions are studied in [4, 11]. Gao and Rouzé [11, Corollary 5.4 and Remark 5.5] also obtain index-independent bounds from mixing in completely positive order; converting L_2 estimates into that mixing condition introduces a conditional-expectation index. Theorem 4.1 instead starts with an entropy contraction coefficient. Its conversion to AT has no additional dimension or index factor.

For $\mathcal{N} \neq B(\mathcal{H})$, the optimal entropy contraction coefficient and the optimal AT coefficient are given by

$$\begin{aligned} \eta_{\text{av}} &= \sup_{\rho \neq \rho_{\mathcal{N}}} \frac{D(\Phi_{\text{av},*}\rho\|\rho_{\mathcal{N}})}{D(\rho\|\rho_{\mathcal{N}})}, \\ c_{\text{AT}} &= \sup_{\rho \neq \rho_{\mathcal{N}}} \frac{D(\rho\|\rho_{\mathcal{N}})}{D(\rho\|\rho_1) + D(\rho\|\rho_2)}. \end{aligned} \quad (4.20)$$

The denominator defining c_{AT} is positive for $\rho \neq \rho_{\mathcal{N}}$: if both defects vanish, both state expectations fix ρ , and convergence of the averaged powers gives $\rho = \rho_{\mathcal{N}}$. Moreover, (4.2) shows that each defect is at most $D_{\mathcal{N}}(\rho)$, so $c_{\text{AT}} \geq 1/2$. The above discussion shows that

$$\frac{1}{2c_{\text{AT}}} \leq 1 - \eta_{\text{av}} \leq \frac{3}{2c_{\text{AT}}}.$$

Here we use the convention $1/(+\infty) = 0$. We can now derive AT from HC for the averaged map.

Corollary 4.2 (HC implies AT). *Let $E_i : B(\mathcal{H}) \rightarrow \mathcal{N}_i$, $i = 1, 2$, be two σ -preserving conditional expectations with trivial intersection $\mathcal{N}_1 \cap \mathcal{N}_2 = \mathbb{C}1$. Suppose that, for some $1 < p < q < \infty$,*

$$\|\Phi_{\text{av}}\|_{p \rightarrow q, \sigma} \leq 1.$$

Then we have

$$D(\Phi_{\text{av},*}\rho||\sigma) \leq \min\left\{\frac{p}{q}, \frac{q'}{p'}\right\} D(\rho||\sigma),$$

where $p' = p/(p-1)$ and $q' = q/(q-1)$. This further implies

$$c_{\text{AT}} \leq \frac{3}{2(1 - \min\{\frac{p}{q}, \frac{q'}{p'}\})}. \quad (4.21)$$

Here we use the fact that the average Φ_{av} is KMS self-adjoint. For each fixed system with scalar intersection, this implies $\|\Phi_{\text{av}} - E_\sigma\|_{2 \rightarrow 2} < 1$. Together with any finite $2 \rightarrow q$ endpoint, Theorem 2.4 also supplies an HC exponent and hence a finite AT constant.

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